

HARRIETTE ADHIAMBO OKAL, PhD

WATER RESOURCES | CLIMATE RESILIENCE | GEOSPATIAL INTELLIGENCE

Hydrology · Water Security · GIS & Earth Observation · Digital Water Systems · Investment Planning

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PROFESSIONAL PROFILE

Water resources and climate resilience specialist bridging hydrological science, geospatial intelligence and development operations to strengthen water-investment decisions across Sub-Saharan Africa. Currently supporting the World Bank Water Global Practice in Africa East across water security, WASH, irrigation, basin planning and climate-resilient investment in Kenya and Somalia.

Combines advanced training in hydrology with applied expertise in climate-risk assessment, GIS and Earth observation, hydroinformatics and digital monitoring. Experienced in translating complex water, climate and spatial evidence into decision-support tools, investment priorities and practical guidance for governments and development partners.

SELECTED SCALE & IMPACT

19

Kenyan counties supported through climate & water investment planning

12

Major African river basins covered through doctoral research

729

Hydrological studies systematically analysed

2 countries

Current World Bank operational portfolio across Kenya & Somalia

CORE EXPERTISE

Water Resources & Water Security: IWRM · basin planning · hydrology · groundwater · WASH/MUS · water governance

Climate Resilience: climate risk & vulnerability assessment · adaptation · drought & flood risk · climate services · NbS

Geospatial Intelligence: GIS · remote sensing · Earth observation · spatial analytics · Google Earth Engine

Digital Water Systems: GIS/MIS · digital monitoring · dashboards · KoboToolbox · decision-support systems

Investment & Operations: water investment planning · project preparation & supervision · technical review · monitoring & learning

Data & Analytics: R · Python · MATLAB · Power BI · hydrological modelling · climate & spatial data analysis

PROFESSIONAL EXPERIENCE

Consultant – Water Resources, Climate Resilience & Hydroinformatics | 2025–Present

World Bank Group, Water Global Practice – Africa East

Support the preparation and implementation of World Bank-financed water, WASH, irrigation and landscape programmes in Kenya and Somalia, with assignments spanning water-resources and climate-risk integration, hydroinformatics and digital monitoring, investment planning, technical quality assurance, government co-design and operational support.

WORLD BANK GROUP – SELECTED PORTFOLIO

Barwaaqo Rural & Urban Water Resilience Programme | Somalia

- Lead development and operationalisation of the programme's Geo-Enabling Monitoring System (GEMS), integrating geotagged project data, spatial analytics and interactive dashboards to strengthen portfolio monitoring, supervision and decision-making.
- Provide Water Resources Management and basin-planning support, integrating surface water, groundwater, climate, environmental and land-use evidence to strengthen catchment analysis and the identification of water-security investment priorities.
- Support State-Level Water and Land Management Planning, developing integrated basin diagnostics and analytical frameworks linking water availability, land degradation, water retention, recharge, Nature-Based Solutions and infrastructure options with investment prioritisation.
- Lead technical work on El Niño preparedness, including spatial risk analysis and monitoring frameworks linking hydroclimatic hazards with programme exposure, potential impacts and preparedness/response actions.

Kenya Resilient Irrigation for a Sustainable Economy (K-RISE/NISIP) | Kenya

- Serve as NIIC Workstream Integrator, providing technical coordination and quality assurance across the Centre's monitoring infrastructure, digital platforms and decision-support tools, capacity development, and data-governance workstreams.
- Maintain coherence across the NIIC technical architecture and support translation of the strategic roadmap into implementable workstream requirements, technical specifications, sequencing and procurement packages.

- Coordinate government co-design and inter-agency engagement across MoWSI, NIA, WRA and other data partners, supporting institutional ownership, user requirements, data governance and the operational foundations of the NIIC.
- Coordinate NIIC requirements across NISIP components and support implementation monitoring, technical risk identification and reporting to the World Bank Task Team and government counterparts.

Kenya Water, Sanitation and Hygiene Programme (K-WASH) | Kenya

- Provide technical review and advisory support on County Water and Sanitation Services Investment Plans (CWSSIPs) and Priority Investment Action Plans (PIAPs), strengthening integration of IWRM, climate-risk screening and water-resource considerations across 19 participating counties.
- Developed a Climate Risk & Water Investment Dashboard, a practical tool for rural water investment, integrating hydrological, groundwater, climate and socioeconomic evidence for spatial risk screening and investment prioritisation.
- Develop climate-proofing screening and retrofitting guidance, alongside MUS/WASH+ decision-support tools, to strengthen the climate resilience and multiple-use functionality of rural water investments.
- Synthesize cross-county findings and support technical clinics, field engagements and operational documentation to translate recurring water-resource and climate-resilience gaps into programme-level learning and guidance.

Financing Locally-Led Climate Action (FLLoCA) | Kenya

- Integrate Water Resources Management and climate-resilient water considerations into county climate-planning workflows, including guidance for embedding WRM and climate-risk screening into investment prioritisation.
- Serve as a technical bridge between FLLoCA and K-WASH, identifying shared methodologies, tools, investment interfaces, implementation risks and opportunities for cross-programme learning.
- Develop practical knowledge-management frameworks, templates and technical products to capture and translate lessons from county planning, field engagements and technical discussions into reusable programme guidance.

Cross-Cutting Water GP Support

- Support project preparation and implementation through technical assessments, analytical work, Implementation Support Missions and implementation follow-up.
- Review ToRs, consultant outputs, technical studies and investment frameworks for methodological quality, climate resilience, technical feasibility and implementation risk.
- Prepare management briefs, technical notes, analytical products and presentations, and coordinate across government institutions, technical agencies, consultants, development partners and multidisciplinary World Bank teams.

Associate Scientist – Water Systems | 2024–2025

Stockholm Environment Institute (SEI), Africa Centre

- Led interdisciplinary Water-Energy-Food Nexus and climate-resilience assessments in Kenya's Ewaso Ng'iro North Catchment, combining hydrological, socioeconomic, governance and geospatial evidence.
- Supported Nature-Based Solutions and ecosystem-based adaptation, including watershed restoration, spring protection and agroforestry.
- Translated water and climate evidence into programme and policy recommendations; developed MEL frameworks, analytical reports, policy briefs and knowledge products.
- Coordinated partnerships with county governments, communities, researchers and development partners.

Doctoral Researcher – Hydrology | 2021–2026

Institute for Water Research, Rhodes University

- Conducted continental research covering 12 major African river basins and systematically analysed 729 hydrological studies.
- Evaluated global hydrological models, reanalysis, Earth Observation and locally derived information for water-resource decision-making in data-scarce environments.
- Examined how data availability, uncertainty and modelling choices shape African water evidence and developed an evidence-triangulation framework for multiple, limited or conflicting information sources.

SELECTED TECHNICAL & PROFESSIONAL ENGAGEMENTS

World Meteorological Organization | Integrated Drought Management Programme & Associated Programme on Flood Management | 2022–Present

Contribute to technical guidance, case studies, Communities of Practice and knowledge exchange on drought preparedness, flood management, climate services and IWRM, as a Water Youth Network Volunteer.

Global Centre for Climate Mobility | Climate Mobility Delegate | 2023–2024

Contributed to African climate-policy dialogue and co-authored the Africa Youth Climate Mobility Declaration.

ETH Zürich – ETH4D | Visiting Researcher & Mentor | 2022–2023

Collaborated on hydrological and water-security research and supported postgraduate researchers with modelling, calibration, validation and uncertainty assessment.

University of Embu | Research Assistant – Land Degradation Neutrality Project | 2020

Applied GIS, remote sensing and field evidence to land-degradation assessment, vegetation analysis and restoration prioritisation.

EWASCO | Assistant GIS Officer – Water Utility Operations | 2015–2016

Applied GIS to water-supply infrastructure planning, asset information and operational decision support.

TECHNICAL CAPABILITIES

Climate & Risk: climate risk and vulnerability assessment · drought and flood risk · El Niño preparedness · climate adaptation · water security · NbS

GIS & Earth Observation: ArcGIS Pro · QGIS · Google Earth Engine · Sentinel · Landsat

Hydroclimatic Information: ERA5-Land · GRACE/GRACE-FO · SMAP · ground and satellite observations

Data & Analytics: R · Python · MATLAB · Power BI · Advanced Excel

Hydrological Modelling: WEAP · Pitman/SPATSIM · SWAT

Digital Monitoring & Decision Support: GEMS · KoboToolbox · dashboards · spatial decision-support products

EDUCATION**PhD in Hydrology | 2021–2026**

Rhodes University, Institute for Water Research, South Africa

MSc, Land & Water Management | 2017–2019

University of Embu, Kenya

BSc, Water Resource Management | 2013–2016

University of Nairobi, Kenya

SELECTED PUBLICATIONS & KNOWLEDGE PRODUCTS

- **Okal, H. A.**, Ngetich, F. K., & Okeyo, J. M. (2020). Spatio-temporal characterisation of droughts using selected indices in Upper Tana River Watershed, Kenya. *Scientific African*, 7, e00275. DOI: [10.1016/j.sciaf.2020.e00275](https://doi.org/10.1016/j.sciaf.2020.e00275)
- **Okal, H. A.**, Ngetich, F. K., & Okeyo, J. M. (2019). Spatial evaluation of droughts using selected satellite-based indices in the Upper Tana River Watershed, Kenya. *International Journal of Plant & Soil Science*, 30(1), 1–13. DOI: [10.9734/IJPSS/2019/v30i130164](https://doi.org/10.9734/IJPSS/2019/v30i130164)
- **Okal, H.**, Tanner, J., & Hughes, D. (2023). Comparison and validation of outputs from global hydrological models against local hydrological data in South Africa. *IAHS Scientific Assembly*. DOI: [10.5194/iahs2022-695](https://doi.org/10.5194/iahs2022-695)
- **Okal, H. A.**, et al. (2024). Balancing Act: Applying the Water-Energy-Food Nexus to Promote Ecosystem-Based Adaptation in the Ewaso Ng'iro North Catchment, Kenya. Stockholm Environment Institute.

Full research profile: [Google Scholar](#)

SELECTED SPEAKING & PROFESSIONAL ENGAGEMENTS

World Bank E4T Webinar | 2026: Invited speaker on climate-resilient water systems and Africa's WASH gap

Drought Resilience +10, Geneva | 2024: Panellist on integrated drought management and climate services

UN Space4Water Conference, San José | 2024: Presenter and moderator on space technologies for water management

WMO/GWP Technical Exchanges: Facilitator and moderator on drought and flood management

RECOGNITION, LANGUAGES & AFFILIATIONS**RECOGNITION**

World Bank Africa Fellow · 2025
ETH4D Doctoral Mentorship Fellow · 2022–2023

LANGUAGES

English · Fluent
Swahili · Fluent
French · Working proficiency
Korean · Intermediate

AFFILIATIONS

IAHS · IWA · IDMP Community of Practice
Water & Climate Coalition · EGU Hydrological Sciences
UNOOSA · Water Youth Network